The Calibration Objective

Vega-normalized residuals, observation weighting, bid–ask bands, and fit quality

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Abstract

Every model in the Vol-Fitter — LQD, SVI, Multi-Core SIV, local volatility — is calibrated by minimizing the *same* kind of objective, and this note documents that shared machinery so the model notes need not repeat it. Three ingredients: a residual *scale* (vega-normalized price, so the loss behaves like a volatility error while every iterate stays a valid price); a per-quote *weight* (equal, or the density-corrected time-value weighting that prevents a dense strike cluster from dominating a sparse but informative wing); and a *target* (the mid, or a bid–ask/haircut band that lets the fit sit anywhere inside the quoted spread and only resists leaving it). We give the weighting formula and its uniform-grid sanity check, the squared-hinge band objective with its small mid anchor and its space-agnostic design, and the goodness-of-fit metric that reports the *actual* calibrated misfit rather than a decorative RMS. Figures generated by the production calibrator show the density correction re-balancing weight from the ATM cluster toward the wings, and a band fit relaxing the pull toward noisy mids.

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1 The residual scale: vega normalization

A smile is quoted in volatility but priced in price, and the two have very different scales across moneyness: a vol bp in the wings is worth a tiny fraction of a vol bp ATM in price terms. The Vol-Fitter resolves this once, for all models, by working in *vega-normalized price* residuals. For a quote at k with total variance w, the residual is

$$r = \frac{C^{\text{model}}(k) - B(k,w)}{\partial_\sigma B(k,w) + \eta}, \qquad \partial_\sigma B = \varphi(d_+)\sqrt{T}, \tag{1}$$

with a small vega floor η in the far wings. The numerator is computed in the model's native, arbitrage-preserving price space (so every iterate is a valid density); the division by vega undoes the price-to-vol scale, so the optimizer sees something close to a uniform volatility error.

Heuristic

Why not just fit volatility directly? Because for the density-based models (LQD, LV) the *price* is the arbitrage-free object and the implied vol is a nonlinear read-off; fitting price keeps the feasible set convex-clean, and the vega division recovers the vol-like loss for free. For the vol-space models (SVI, SIV) the residual is already a vol error and (1) degenerates to exactly that. One residual concept, two implementations, identical units.

2 Observation weighting

2.1 The problem with counting quotes

Each squared residual enters the loss with a weight ω_i. With equal weights the fit is driven by wherever the quotes happen to be *dense* — typically a tight ATM cluster — so an isolated but economically important wing quote barely registers. The desk wants the opposite: weight should follow the *economic* importance of a strike, not the accident of how finely it was sampled.

2.2 Density-corrected time-value weights

The Vol-Fitter offers two schemes via `weightScheme`. The default `equal` uses unit weights. The `tv_density` scheme implements

$$\boxed{\omega_i = \max(\mathrm{TV}_i, \varepsilon)\,\frac{s_i}{\bar{s}},} \tag{2}$$

where TV_i is the quote's *time value* (its OTM option price, which in forward-normalized units is its whole value), and s_i is the one-dimensional Voronoi cell width of the quote in log-strike — half the gap to each neighbour, one-sided at the ends. The factor TV_i is the economic weight (a quote on a richly-priced option matters more); the factor $s_i/\bar{s}$ is the *density correction*, dividing out the local sampling density $1/s_i$ so that the *aggregate* weight over strike space follows the time-value measure $\mathrm{TV}(x)\,\mathrm{d}x$ rather than the raw quote histogram.

Proposition 1 (Uniform-grid sanity check)**.** *On a uniform log-strike grid all s_i are equal, so $s_i/\bar{s} = 1$ and (2) reduces to $\omega_i = \mathrm{TV}_i$ — the pure time-value weighting, with no double-counting to correct.*

A cap `DEFAULT_MAX_MULT`$= 10$ on the spacing multiplier $s_i/\bar{s}$ stops a single isolated far-wing quote from dominating. Crucially the weights are *mean-normalized to one*, so switching schemes leaves the data-versus-regularization balance unchanged: the LQD ridge, the sigmoid hat ridge and the LV roughness penalty are all tuned against unit-mean weights and never silently over- or under-regularize when the scheme changes.

Figure 1 shows the effect on a non-uniform strike set with a dense ATM region and sparse wings: the dense ATM quotes are individually *down*-weighted and the sparse wing quotes *up*-weighted relative to equal, re-balancing the fit toward the full strike range while still respecting that ATM options carry more time value (the net wing-to-ATM weight ratio here is 1.3).

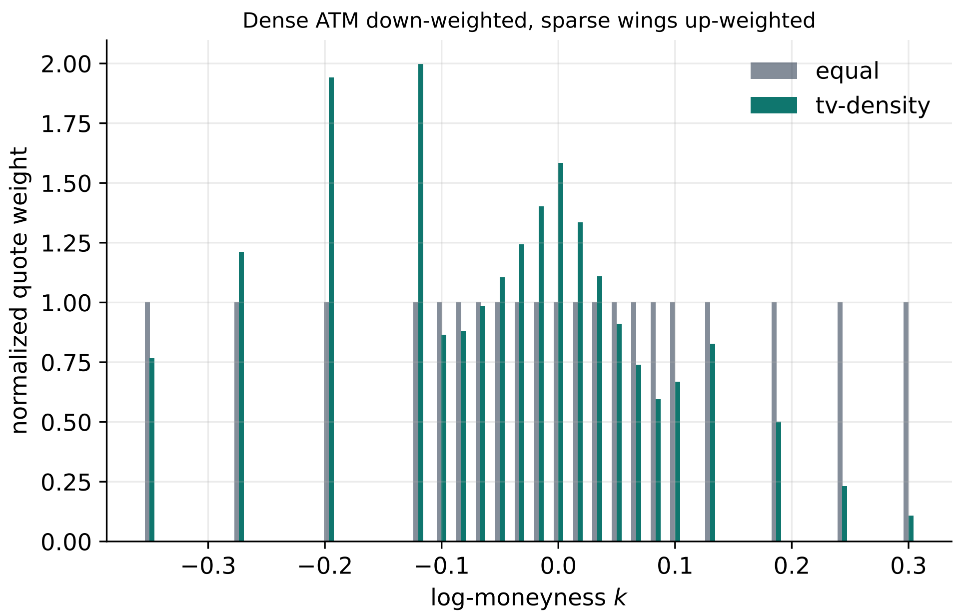

Figure 1: Equal vs density-corrected time-value weights on a non-uniform strike set. The dense ATM cluster is down-weighted per quote, the sparse wings up-weighted, so the aggregate weight follows the time-value measure rather than the quote histogram.

3 Bid–ask bands

3.1 Why a band, not a mid

A mid quote pretends a price is known exactly; a real quote is a *band* $[\text{bid}, \text{ask}]$, and any curve inside it is consistent with the market. Fitting the mid forces the curve to chase the midpoint of every spread, including the noise in illiquid wings where the spread is wide. The band objective instead penalizes only *leaving* the band, with a gentle pull toward mid for identifiability:

$$\boxed{\text{loss}_i = \max(m_i - \text{hi}_i, 0)^2 + \max(\text{lo}_i - m_i, 0)^2 + \theta_{\text{mid}}\,(m_i - \text{mid}_i)^2,} \tag{3}$$

where m_i is the model value, $[\text{lo}_i, \text{hi}_i]$ the (possibly haircut-adjusted) band, and $\theta_{\text{mid}} = 0.05$ the mid-anchor weight. Inside the band the first two terms vanish: the fit is *free*, paying only the small anchor; once it leaves, the squared hinge pulls it back hard.

3.2 Modes and the haircut

Three modes via `fitMode`. `mid` sets lo = hi = mid (the band collapses, recovering a mid fit). `bidask` uses the raw $(\mathrm{bid}, \mathrm{ask})$. `haircut` tightens each side toward mid by `haircut` volatility points, never crossing mid,

$$\mathrm{lo} = \min(\mathrm{bid} + h, \mathrm{mid}), \qquad \mathrm{hi} = \max(\mathrm{mid}, \mathrm{ask} - h), \tag{4}$$

so a quote tighter than $2h$ collapses gracefully to a mid fit on that strike. The hinge is monotone in the quote value, so the *same* construction works in any monotone residual space — implied vol for SVI/SIV, vega-normalized price for LQD/LV — and the band-violation subgradient sgn feeds the analytic Jacobians. Both are one line:

Listing 1: The two-sided band hinge and its subgradient (distilled from `calib/band.py`; match production exactly).

```python
def band_violation(model, lo, hi):    # > 0 only outside [lo, hi]; exactly 0 inside
    return np.maximum(model - hi, 0.0) + np.maximum(lo - model, 0.0)

def band_sign(model, lo, hi):         # d(violation)/d(model): +1 above, -1 below, 0 inside
    return np.where(model > hi, 1.0, 0.0) - np.where(model < lo, 1.0, 0.0)
```

Figure 2 shows a band fit against a mid fit on a smile with a widening wing spread and noisy mids: the band fit is content to sit anywhere inside the shaded band, relaxing the pull toward the noisy midpoints, while the mid fit is anchored to each midpoint.

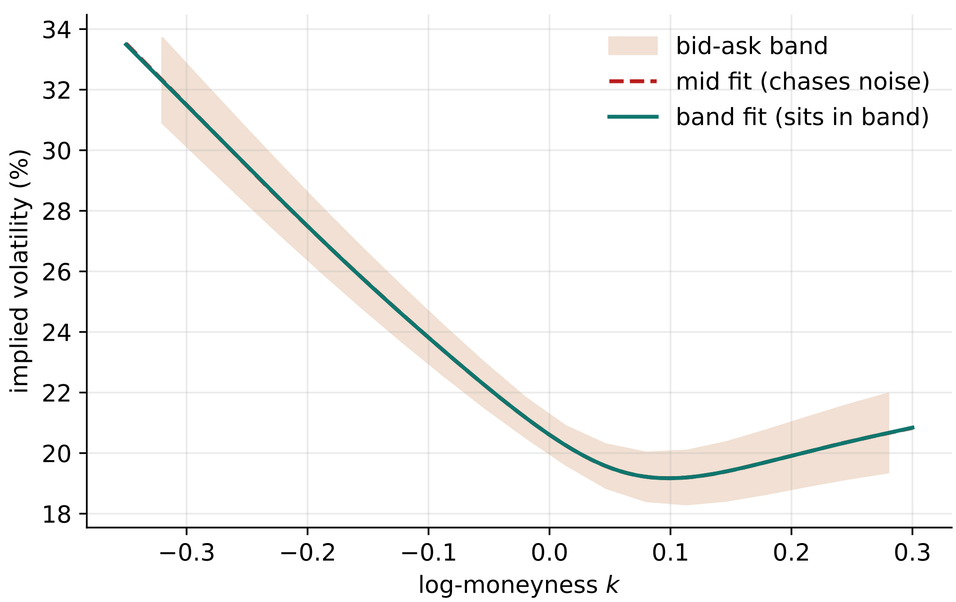

Figure 2: A bid–ask band fit (solid) vs a mid fit (dashed) with the quoted band shaded. Inside the band the fit pays only the small mid anchor, so it need not chase noisy midpoints — most valuable in wide-spread wings.

Heuristic

The band's value scales with model flexibility. A rigid parametric model (a seven-parameter LQD) already smooths through mid noise, so the band and mid fits nearly coincide; a flexible model (a many-vertex local-vol surface) can be pulled into spurious wiggles by noisy mids, and there the band's freedom-to-sit-anywhere is what keeps the surface smooth. The band is a *regularizer that costs no bias*: it does not push the fit anywhere, it merely stops penalizing it for being inside the spread.

4 Goodness of fit

A reported fit error should reflect the *objective that was minimized*, not a decorative quantity. The Vol-Fitter's weighted RMS therefore mirrors the active configuration: the distance to the fit *target* (mid, or band violation, so a fit sitting happily inside the band reports near-zero error even if it is far from mid), under the active *weighting* scheme, including the var-swap penalty term when active. It is returned as a $(\sum \omega r^2, \sum \omega)$ pair so node errors aggregate correctly into a surface RMS. This is why the same surface can report a different RMS under `mid` and `bidask`: the two minimize different objectives, and the metric is honest about which.

5 What is original here

The vega normalization is standard; the contributions are the *density-corrected* weighting (2) — separating economic importance (time value) from sampling density (Voronoi spacing), with the mean-normalization that keeps regularization invariant across schemes — and the *space-agnostic squared-hinge band* (3) whose monotone subgradient plugs into every model's analytic Jacobian, so bid–ask fitting is a first-class mode rather than a bolt-on. Together they let the desk express "fit what matters, to within the spread" in two orthogonal knobs.

A Hyperparameter atlas

Table 1: Calibration-objective hyperparameters.

Knob	Default	Role
Surfaced (FitSettings / OptionsSettings)		
`weightScheme`	`equal`	`equal` or `tv_density`.
`fitMode`	`mid`	`mid` / `bidask` / `haircut`.
`haircut` h	0.005	Haircut in vol points (each band side toward mid).
`midAnchorWeight` θ_{mid}	0.05	Mid anchor strength vs band penalty (= 1).
Hidden		
`DEFAULT_MAX_MULT`	10	Cap on the Voronoi spacing multiplier $s_i/\bar{s}$.
ε	10^{-12}	Floor on time value / spacing.
η (vega floor)	10^{-4}	Floor on Black vega in (1).

Performance. Weighting is $O(m \log m)$ (one sort) and the band residual is $O(m)$, both negligible against the fit; the band subgradient is analytic so it adds nothing to Jacobian assembly. Switching schemes or modes bumps the options version and triggers exactly one refit.

B Reference implementation: the band residual stack

The full objective stacks two blocks per quote: the band violation (scaled by the per-quote weight, or 1/vega in the LQD/LV price space) and a weak mid anchor of strength θ_{mid}, so the fit is free inside the band yet never drifts off a one-sided quote. Squaring and summing reproduces the loss $\max(m - \mathrm{hi}, 0)^2 + \max(\mathrm{lo} - m, 0)^2 + \theta_{\mathrm{mid}}(m - \mathrm{mid})^2$. Matches production to 10^{-15}.

Listing 2: The stacked band-objective residual (distilled from `calib/band.py`).

```
def band_residuals(model, lo, hi, mid, scale, mid_anchor_weight):
    viol = scale * band_violation(model, lo, hi)          # push inside the bid/ask band
    anchor = np.sqrt(mid_anchor_weight) * scale * (model - mid)   # weak pull to mid
    return np.concatenate([viol, anchor])                 # least-squares stack (length 2N)
```

References

[1] Vol-Fitter Technical Note, *Density-corrected time-value weights* (`Docs/iv_time_value_density_weights.tex`).

[2] J. Gatheral. *The Volatility Surface*. Wiley, 2006.